

Systems Under Pressure

Investigation Notebook

DRIVING QUESTION

How do we know when a system is actually in danger?

GROUP NAME

MEMBERS

INVESTIGATION TOPIC

CATEGORY

ESSENTIAL QUESTION

NOTEBOOK STRUCTURE

Pages 1–3 · Empathy + Explore

Pages 4–6 · Systems + Intervals

Pages 7–9 · Build Models

Pages 10–12 · Critique + Revise

Pages 13–15 · Claim + Present

HOW TO USE THIS NOTEBOOK

This is a field journal — not a worksheet.

Write, sketch, map, and calculate anywhere on the page.

Show your thinking — including what changed.

Claim → Evidence → Test → Revision

Vlogs are formative checkpoints (4–6 min each):

record after each class and upload with your
individual artifact and written reflection.

Written reflection: minimum 150 words per vlog.

Start here — before categories, before models, before data. Understand the situation first.

AI CO-CREATION PROMPTS

- ## WHO IS AFFECTED — AND WHAT DO THEY EXPERIENCE?

WHO HOLDS POWER OVER THIS SYSTEM — AND WHAT DO THEY PRIORITISE?

WHAT CONFUSES OR SURPRISES US

WHAT MAKES THIS URGENT / INTERESTING

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Before building models, understand who is affected and how. Use AI to gather perspectives.

AI CO-CREATION PROMPTS

- ## STAKEHOLDER 1 — WHO THEY ARE + WHAT THEY EXPERIENCE

Variables + Hidden Systems

Map connected variables and systems. Think in pressures, delays, and ripple effects.
A confounding variable is a third factor that could explain what you think you see. Name at least one.

AI CO-CREATION PROMPTS

- "What variables are connected to [your topic] that are rarely measured?"
- "If [variable A] increases, what else might change — and with what time delay?"
- "What confounding variable would a skeptic point to if I claimed [A] causes [B]?"

CONNECTED SYSTEMS MAP — DRAW ARROWS, PRESSURES, CONNECTIONS, DELAYED EFFECTS

VARIABLES WE CAN MEASURE

VARIABLES WE CANNOT EASILY MEASURE (HIDDEN)

REQUIRED SENTENCE FRAME

These variables appear related because ,
but this does not prove causation because .

For each source record: variable measured · timeframe · credibility · limitations · keep or reject.
Evaluate methodology, not just credibility. At least one source kept and one rejected with explicit reasons.

[illegible]

SOURCE I QUESTIONED OR REJECTED — WHY

Slow down. Annotate before interpreting. A threshold is the value at which behavior changes. A turning point is where direction reverses. Name both if present. What does the graph NOT show?

paste, sketch, or annotate here

ANOMALIES OR SURPRISES

Local vs Global Behavior

Local behavior: what happens in a short interval. Global behavior: the long-term pattern.
Choosing an interval is a mathematical decision with real consequences. Which interval is most honest?

SHORT INTERVAL — LOCAL BEHAVIOR

sketch

LONG INTERVAL — GLOBAL BEHAVIOR

sketch

WHAT CHANGES WHEN I ZOOM OUT? WHAT PATTERNS ONLY APPEAR AT A LONGER SCALE?

WHICH INTERVAL TELLS THE MOST HONEST STORY — AND WHAT ARE THE CONSEQUENCES OF CHOOSING IT?

Investigate repeated change. Build a ratio/difference table. Is the common difference constant (arithmetic) or the common ratio constant (geometric)? Write both recursive and explicit (closed-form) formulas. Calculate a partial sum S_n . Identify where the pattern becomes unreliable.

[illegible]

PARTIAL SUM $S_n = A_1(1-R^n)/(1-R)$ — REAL-WORLD TOTAL

Exponentials + Thresholds

Test whether the common ratio a/a_{-1} is stable enough for $f(x) = a \cdot b^x$.
Solve for threshold using $\log_b(\text{threshold}/a) = x$. Distinguish continuous $f(x)=a \cdot b^x$ from discrete $a_n = a_1 \cdot r^{n-1}$. Where does the model break down?

x	y (value)	y / y ₋₁ (ratio)	Ratio stable?	Notes

EXPONENTIAL MODEL $F(x) = A \cdot B^x$ — WHAT DO A AND B MEAN IN CONTEXT?

CONTINUOUS $F(x)=A \cdot B^x$ OR DISCRETE $A_n=A_1 \cdot R^{n-1}$? JUSTIFY.

THRESHOLD SOLVE — $\log_b(\text{THRESHOLD} / A) = X$

$\log_b(\text{threshold} / a) = x \rightarrow x =$ _____

In context this means: _____

WHERE DOES EXPONENTIAL REASONING FAIL OR BECOME MISLEADING FOR THIS SYSTEM?

Polynomial Behavior

Focus on graph shape over a chosen local interval. Identify turning points, zeros, and multiplicity. Even multiplicity → graph bounces (tangent). Odd multiplicity → graph crosses. Connect each feature to a real threshold.

GRAPH SKETCH — LOCAL INTERVAL

sketch graph here

FEATURE ANNOTATIONS

Zeros / threshold crossings:

Turning point(s):

Multiplicity (bounce or cross):

Local max / min value:

Interval chosen:

End behavior (and why unrealistic):

FACTORED FORM + INTERPRETATION

$P(t) = a(t - r_1)(t - r_2)^2 \dots$

WHAT DOES $P(T) = A(T-R_1)(T-R_2)^2 \dots$ REVEAL THAT $F(X) = A \cdot B^x$ OR A LINEAR MODEL CANNOT?

Model Comparison

Compare all model types for the same data. Each reveals something. Each hides something.
Fill every cell — then choose the most defensible model and explain why the others are less appropriate.

CRITERION	Linear	Geometric / Exponential	Polynomial P(t)	Arithmetic Sequence
Equation / Form				
What it reveals				
What it hides				
Main assumption				
Domain / valid range				
Fails when...				

MOST DEFENSIBLE MODEL — AND WHY THE OTHERS ARE LESS APPROPRIATE FOR THIS SYSTEM

Confounding Variables

A confounding variable is a third variable that could explain the relationship between your two main variables.
Name each one, explain its effect, and use the required sentence frame. Without this, correlation cannot support any causal claim.

CONFOUNDING VARIABLE 1 — Name: _____

CONFOUNDING VARIABLE 2 — Name: _____

CONFOUNDING VARIABLE 3 — Name: _____

CONFOUNDING VARIABLE 4 — Name: _____

REQUIRED SENTENCE FRAME

These variables appear related because _____ ,
but this does not prove causation because _____ .

Flint / AI Critique Log

Document every meaningful use of Flint. This is analytical evidence of your critical thinking — not a compliance log.
Show what Flint got wrong or what you rejected and why. Protocol: Think First → Flint → Verify → Decide.

THINK FIRST → FLINT → VERIFY → DECIDE

ENTRY 1

What I asked Flint (the prompt):

What Flint suggested:

What I accepted — and why:

What I rejected or corrected — and why:

How I verified independently:

ENTRY 2

What I asked Flint (the prompt):

What Flint suggested:

What I accepted — and why:

What I rejected or corrected — and why:

How I verified independently:

ENTRY 3

What I asked Flint (the prompt):

What Flint suggested:

What I accepted — and why:

What I rejected or corrected — and why:

How I verified independently:

GROUP

DATE

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Make changing thinking visible. Every significant revision should be traceable to a specific piece of evidence or contradiction. Bigger space = more cognitive work expected. The revised interpretation is the most demanding section.

[illegible][illegible]

Final Claim

What is your most responsible mathematical interpretation? Be specific. Name your model, evidence, limitation, and uncertainty.
Use hedged language: "the data is consistent with," "our model suggests," "we cannot rule out." Honest uncertainty is strength.

OUR STRONGEST MODEL — NAME IT AND EXPLAIN WHAT IT CAPTURES THAT OTHERS CANNOT

OUR STRONGEST EVIDENCE — THE DATA FEATURE HARDEST TO EXPLAIN AWAY

OUR BIGGEST LIMITATION — WHAT THE MODEL CANNOT CAPTURE OR OVERSIMPLIFIES

REMAINING UNCERTAINTY — WHAT WE BELIEVE BUT CANNOT FULLY PROVE

OUR FINAL MATHEMATICAL CLAIM:

Presentation Prep

Plan your mathematical argument — not a summary of what you did. Know your strongest evidence and your weakest assumption.
Pre-recorded teams: submit video before presentation day. Live teams: be ready for a follow-up question.

OUR CENTRAL CLAIM — ONE SENTENCE, SPECIFIC AND DEFENSIBLE

THE GRAPH WE LEAD WITH — AND WHY IT MAKES THE STRONGEST CASE FOR OUR CLAIM

TWO MODELS WE WILL COMPARE — EQUATIONS, WHAT EACH REVEALS, WHAT EACH HIDES

OUR THRESHOLD — HOW WE CALCULATED $\log_b(\text{THRESHOLD}/A)=X$ AND WHAT X MEANS

OUR BIGGEST LIMITATION — WE WILL NAME THIS BEFORE THEY ASK

FOLLOW-UP QUESTION WE EXPECT — AND OUR EVIDENCE-BASED RESPONSE